

Function Call Diagnostic Test: Solutions

The results do not count for anything. This test is to allow the ECS401 team to diagnose if you need help to ensure they do not fall behind in learning to program. Understanding how for loops work is critical so if you have any problems we want to sort them out now. You must show all your working in the following. No correct table (exactly the same)... no marks.

1. What is printed after a method call `squarePrint()` ;

```
public static void squarePrint ()
{
    int answer = square(3);           // L1
    System.out.println(answer);       // L2
    answer = square(2);               // L3
    System.out.println(answer);       // L4

    return;
}

public static int square (int s)      // L5
{
    int res = s * s;                  // L6
    return res;                       // L7
}
```

Complete the dry run tables overleaf (you may not need all rows or all tables):

squarePrint ()				
LINE	answer	square(3) =?	square(2) =?	printed
L1	9	square(3) =? 9		
L2				9
L3	4		square(2) =? 4	
L4				4

square (3)			
LINE	s	res	return res
L5	3		
L6		9	
L7			return 9

square (2)			
LINE	s	res	return res
L5	2		
L6		4	
L7			return 4

Printed

9

4

2. What is printed after a method call `squarePrint2()`;

```
public static int squarePrint2 ()
{
    int x = 2;                // L1
    x = squarePlus1(x);       // L2
    x = squarePlus1(x);       // L3
    System.out.println(x);    // L4

    return x;                 // L5
}

public static int squarePlus1 (int s)    // L6
{
    int sp1 = (s * s) + 1;               // L7
    return sp1;                          // L8
}
```

Complete the dry run tables (you may not need all rows or all tables):

squarePrint2 ()				
LINE	x	squarePlus1(x) =?	return x	printed
L1	2			
L2	5	squarePlus1(2) =? 5		
L3	26	squarePlus1(5) =? 26		
L4				26
L5			return 26	

squarePlus1 (2)			
LINE	s	sp1	return sp1
L6	2		
L7		5	
L8			return 5

squarePlus1 (5)			
LINE	s	sp1	return sp1
L6	5		
L7		26	
L8			return 26

Printed:
26

3. What is printed after a method call `x6()` ;

```
public static void x6 ()
{
    String s = "2";           // L1
    String t = trip(s);       // L2
    String d = doub(t);       // L3
    System.out.println(t + ":" + d); // L4
    return;
}

public static String doub (String d) // L5
{
    String r = (d + d);       // L6
    return r;                 // L7
}

public static String trip (String t) // L8
{
    String r = (t + t + t);    // L9
    return r;                 // L10
}
```

Complete the dry run tables overleaf (you may not need all rows or all tables):

x6 ()						
LINE	s	t	d	trip(s) =?	doub(d)=?	printed
L1	"2"					
L2		"222"		trip("2")=? "222"		
L3			"222222"		doub("222")=? "222222"	
L4						222:222222

trip ("2")			
LINE	t	r	return r
L5	"2"		
L6		"222"	
L7			return "222"

doub ("222")			
LINE	d	r	return r
L5	"222"		
L6		"222222"	
L7			return "222222"

Printed:
222:222222

4. What is printed after a method call `f()` ;

```
public static void f ()
{
    String s1 = "b";           // L1
    String s2 = "a";           // L1
    String s3 = comb(s2, s1);   // L2
    System.out.println(s2 + s1 + "-" + s3); // L4
    return;
}

public static String comb (String s1, String s2) // L5
{
    String r1 = (s1 + s2);      // L6
    String r2 = (s2 + s1);      // L7
    System.out.println(r2);     // L8
    return (r1 + r2);           // L9
}
```

Complete the dry run tables overleaf (you may not need all rows or all tables):

f ()					
LINE	s1	s2	s3	comb(s2,s1) =?	printed
L1	"b"				
L2		"a"			
L3			"abba"	comb("a","b") =? "abba"	
L4					ab-abba

comb ("a","b")						
LINE	s1	s2	r1	r2	return (r1+r2)	printed
L5	"a"	"b"				
L6			"ab"			
L7				"ba"		
L8						ba
L9					return "abba"	

Printed:

ba
ab-abba